Project 2b – Bayesian Optimization of Ad-Atom Energy Landscapes

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1 Introduction

Despite our never-ending love for physics, we can not deny the mental fatigue that comes with simulating and understanding complex systems. As the systems grow in complexity, so does the computational cost of simulating them. This is especially true in the field of material science, where we often want to understand the properties of materials at the atomic scale. To overcome this challenge, we can use surrogate models that approximate the behavior of the system based on a limited number of simulations. One such approach is Bayesian Optimization (BO), which combines probabilistic modeling with optimization techniques to efficiently explore the parameter space. In this project, we apply BO to model the potential energy surface (PES) of an ad-atom on a gold-surface. By using Gaussian Processes (GPs) as our surrogate model, we aim to predict the energy of the ad-atom at unobserved locations on the surface, while minimizing the number of expensive energy calculations required.

2 Theory

For an ad-atom on a surface, its energy E can be expressed as

$$E = E_{\text{ad}} - E_{\text{surface}}, \tag{1}$$

where E_{surface} is the energy of the bare surface and E_{ad} is the energy of the combined system [1]. If we consider the surface to be completely rigid, the energy landscape of the ad-atom can be described as a function of its position (x, y, z) on the surface. By fixing the height z to its local minimum given (x, y), we can reduce the problem to a two-dimensional energy landscape $E(x, y)$. By combining GPs with BO, we can efficiently model this energy landscape and predict the energy at unobserved locations, while minimizing the number of expensive energy calculations required.

3 Methodology

In this project, we consider a gold (Au) structure with Miller indices (433) and an ad-atom of gold placed on its surface. The energy of the ad-atom is calculated using the Atomic Simulation Environment (ASE) package using the Effective Medium Theory (EMT) potential [2]. As outlined in the introduction, we model the energy landscape (PES) of the ad-atom as a function of its position (x, y) on the surface, while keeping its height z fixed at the local minimum for each (x, y) pair, i.e.,

$$E(x, y) = \min_z E(x, y, z). \tag{2}$$

The minimization, or relaxation, over z was performed using the BFGS algorithm implemented in ASE. Furthermore, when searching for the optimal ad-atom position, the energy landscape will repeat itself due to the periodicity of the crystal structure. Therefore, we only need to consider a single unit cell of the surface, i.e., $0 < x\,(\text{Å}) < 16.65653$ and $0 < y\,(\text{Å}) < 2.884996$. We used a step size of 0.02 Å in both x and y, resulting in a 145×833 grid of possible ad-atom positions. For an initial picture of the energy landscape, it was plotted over the entire grid. From this, the approximate position of the global minimum could be identified for further comparison and analysis. Next, we applied local search using `SciPy`, specifically L-BFGS-B, to investigate the performance of regular gradient-based optimization on this energy landscape [3]. The algorithm was initialized at 500 random positions within the unit cell, and the number of times it converged to the global minimum was recorded.

Next, Bayesian Optimization (BO) with Gaussian Processes (GPs) was employed to model the energy landscape and search for the global minimum. The GP utilized a Radial Basis Function (RBF) kernel combined with a bias kernel to capture the underlying structure of the energy landscape, implemented using `GPy` [4]. From analyzing the energy landscape, local extremes were typically separated by approximately 1–2 Å in both x and y. Therefore, the length scale hyperparameter of the RBF kernel, which represents the characteristic width of features in the energy landscape, was initialized to a Gamma prior with shape 2.0 and scale 1.0, see Figure 1 below [5]. The variance hyperparameter, which controls the overall variability of the function, was initialized to a Gamma prior with shape 2.0 and scale equal to half the variance of the grid energies, σ^2_{PES}. For the bias, the variance hyperparameter was initialized to a Gamma prior with mean equal to one-tenth of the variance of the grid energies to account for any constant offset in the energy landscape but, with less influence than the RBF kernel.

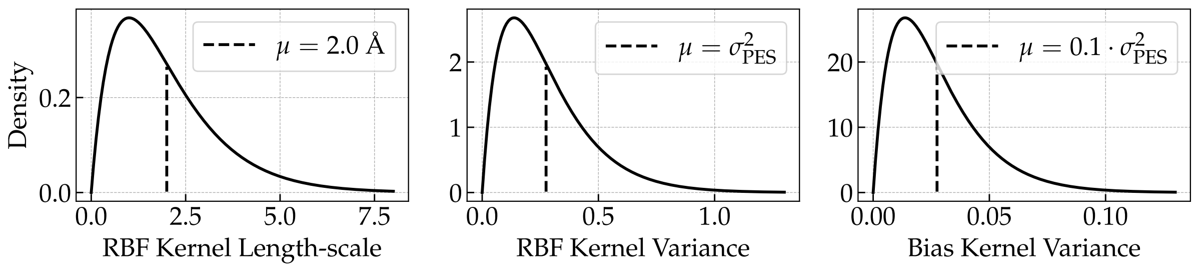

Figure 1: Priors for the length scale and variance hyperparameters of the RBF kernel and the variance hyperparameter of the bias kernel used in the GP model.

For each BO iteration, the Lower Confidence Bound (LCB) acquisition function was employed to select the next sampling point, defined as

$$\text{LCB} \equiv A(x, y) = -\mu(x, y) + \beta\sigma(x, y), \tag{3}$$

where $\mu(x, y)$ and $\sigma(x, y)$ are the predicted mean and standard deviation of the GP model at position (x, y), respectively, and β is a hyperparameter that balances exploration and exploitation. Given this definition, the next sampling point is chosen as the position that maximizes $A(x, y)$. Initially, five randomly drawn samples from the (x, y) domain were used for training. Regarding β, different values were tested to observe their effect on the optimization process and convergence performance. The BO process was run for a total of 50 iterations for each β value, and the energy of the sampled positions at each iteration was recorded. After completing the BO runs, the final GP model was evaluated on the full energy landscape grid using the optimal β.

Finally, a linear transition path between the global minimum and a local minimum located at $(11, 2.1)$ was defined. This was done using the parametrization

$$\boldsymbol{r}(\lambda) = \boldsymbol{r}_{\text{start}} + \lambda(\boldsymbol{r}_{\text{end}} - \boldsymbol{r}_{\text{start}}), \quad \lambda \in [0, 1], \tag{4}$$

where $\boldsymbol{r}_{\text{start}}$ and $\boldsymbol{r}_{\text{end}}$ are the position vectors of the global and local minima, respectively. The energy along this path was computed using both the BO-GP model and a general GP model trained on a larger set of 200 samples from the grid, using seven initial samples, including $\boldsymbol{r}_{\text{start}}$ and $\boldsymbol{r}_{\text{end}}$. To ensure a general model, β was set to $\beta = 10^6$ in order to prioritize exploration during sampling. The same kernel and hyperparameter priors as in the BO-GP model were used. For the latter model, the root mean square error (RMSE) was calculated as a function of the number of training samples to evaluate its performance. Ultimately, the energy profiles along the transition path from both models were compared to the target energy profile computed using EMT.

4 Results

The left panel of Figure 2 shows the potential energy surface $E(x, y)$ of the Au ad-atom evaluated on the 145×833 grid over the primitive cell. The energy varies between roughly 7.05 and 8.15 eV, with several local minima and maxima in a reoccurring pattern. The pattern breaks for $x \lesssim 5$, where the global extremes appear. The global minima is located at $(3.26, 0.72)$ with an energy of $E_{\text{min}} = 7.08\,$eV. The region around the global minimum is rather small as it is narrow in x and moderately extended in y. The right panel of Figure 2 shows 15 trajectories of the local optimization algorithm initialized at 500 random positions over the PES grid. The squares mark the starting positions and the diamonds mark the end positions of the optimizations. Out of the 500 runs, only 67 converged to the global minimum, indicating that regular gradient-based optimization struggles to find the global minimum from most starting positions due to the complex landscape with many local minima. Considering that the grid presents approximately ≈ 15 minima, this result is not unexpected as random starting positions are likely to be closer to other local minima.

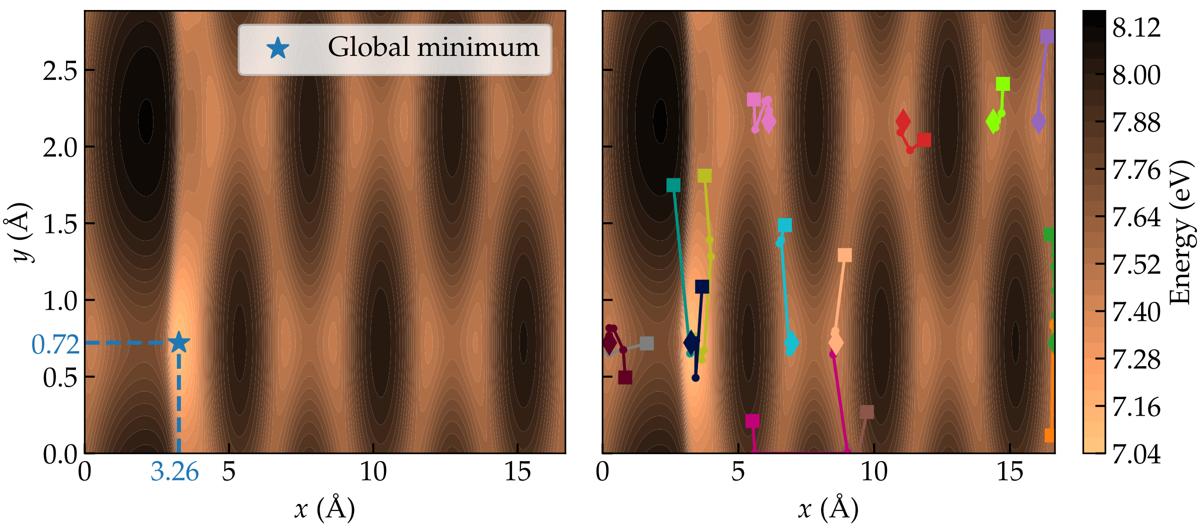

Figure 2: Left: Potential energy surface of the Au ad-atom evaluated on the 145×833 grid over the primitive cell. Right: 15 trajectories of the L-BFGS-B local optimization algorithm initialized at 500 random positions over the PES grid. The squares mark the starting positions and the diamonds mark the end positions of the optimizations.

Figure 3 shows the energy of the sampled position (x, y) at each BO iteration for different β. The dashed horizontal line marks the global minimum energy obtained from the full PES. For $\beta = 0.5$, the sampled energy settles at a higher value and remains there, indicating that the optimization has become trapped in a local minimum. For $\beta = 1$ and $\beta = 2$, the sampled energy reaches the global minimum after about 15 iterations and then stays at this level for the remainder of the run, indicating rapid and stable convergence. For $\beta = 3$, the optimization reaches the global minima several times and stays there for a few iterations before moving away again, not fully converging. For $\beta = 5$, the global minimum is visited once but the sampled energy increases again and does not return to this level within the 50 iterations, while it is not reached at all for $\beta = 10$. Moving forward, $\beta = 2$ was chosen as the optimal value for further analysis due to its reliable, and fast, convergence to the global minimum. Nevertheless, $\beta = 1$ also performed well, but due to the stochastic nature of the BO process, $\beta = 2$ was preferred as it provides a better balance between exploration and exploitation.

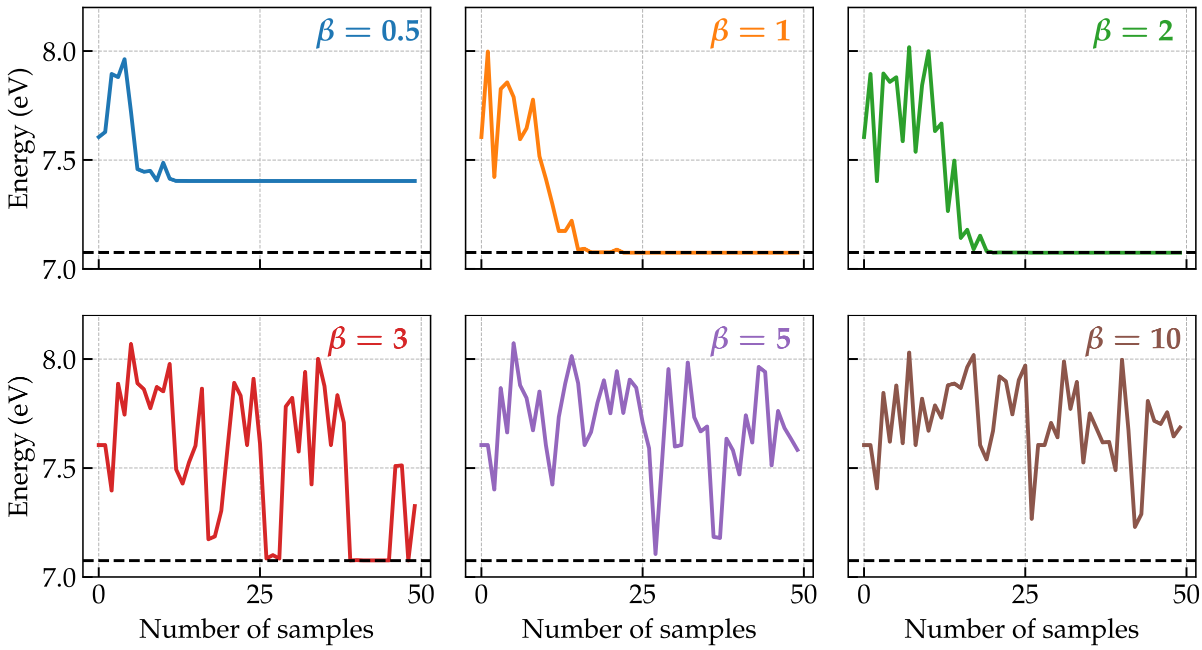

Figure 3: The energy of the sampled position (x, y) at each BO iteration for different values of β. The dashed horizontal line marks the global minimum energy.

Figure 4 illustrates the final BO-GP model for the optimal $\beta = 2$, chosen from Figure 3. The leftmost panel shows the predicted mean energy over the domain together with starting and sampled positions, and the known global minimum. Most sampled positions lie in low energy regions, with a dense cluster in the well that contains the global minimum. The middle panel shows the predictive standard deviation (uncertainty) σ with the same data points. The uncertainty is small in areas that contain many sampled positions, generally around the minimas (especially the global minimum), and larger in regions that have not been visited. The right panel displays the value of the LCB acquisition function for the same model. It gives a low value for the sampled positions at local maxima and high values for the sampled positions at the minima, opposite to the predicted energy as expected.

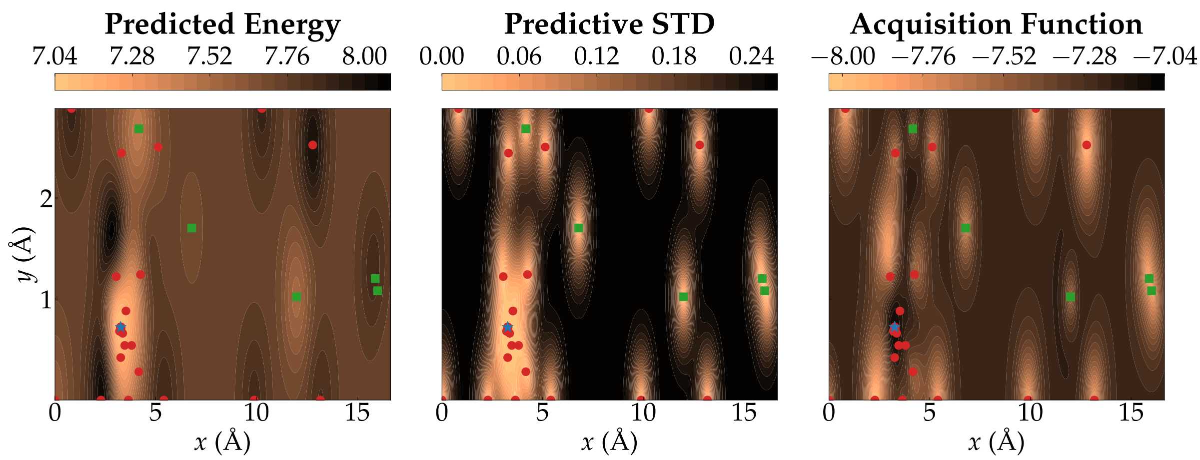

Figure 4: From left to right: Predicted mean energy of the final BO-GP model, predictive standard deviation, and LCB acquisition function values with sampled positions over the domain. Green squares mark the initial samples, red circles mark the samples added during BO, and the blue star marks the global minimum.

The left panel of Figure 5 shows the RMSE of the general GP model evaluated on the full PES as a function of the number of training samples. The RMSE starts at almost $0.4\,\text{eV}$ for very small training sets and decreases rapidly as more samples are added during the first ~ 60 samples. After this point, the decrease is slow and approximately linear. The right panel compares the energy along the linear transition path between the global minimum ($\lambda = 0$) and the local minimum ($\lambda = 1$) at $(11, 2.1)$ between the BO and general GP with the target energy along the path computed with EMT. The BO-GP model starts identically to the target with basically no uncertainty around the global minimum, but then smoothes the rest of path, staying decently stable just under $7.8\,\text{eV}$ with a large uncertainty encompassing the entire target trajectory. The general GP is slightly worse at the global minimum but keeps very close to the target with small uncertainty throughout the path, being a lot better overall.

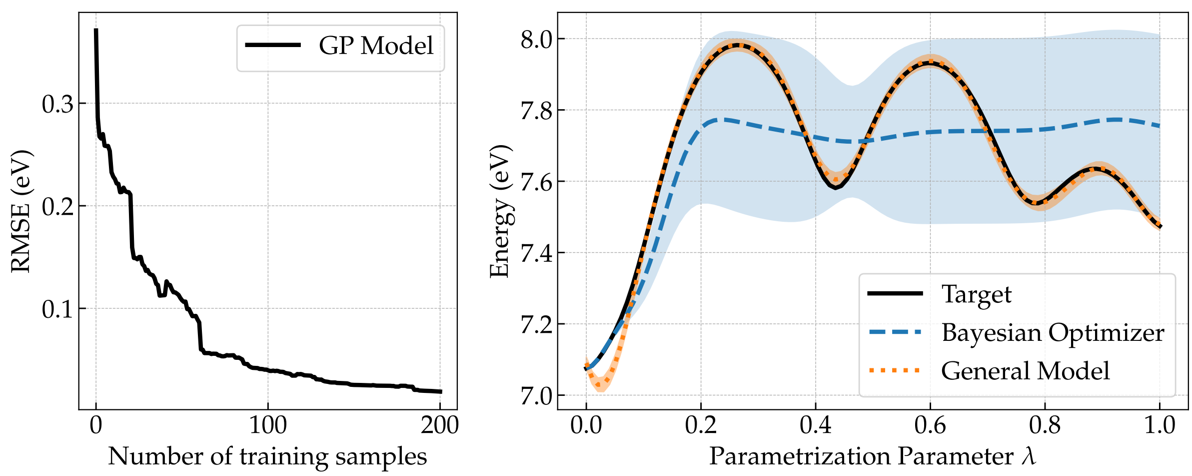

Figure 5: Left: RMSE for the general model as a function of the number of training samples. Right: Comparison of the energy along the linear transition path between the global minimum and the local energy minimum around $(11, 2.1)$ for the BO-GP model, the general GP model, and the target energy computed with EMT.

5 Discussion

The appearance of the PES in Figure 2 explains why the optimization problem can be demanding. Many narrow minima are spread over the unit cell and the global minimum does not stand out clearly by shape alone. Only a small fraction of the domain lies close to the global minimum, which means that a method that only follows local gradients is expected to reach it from relatively few starting positions. This is reflected by the results from the local optimization runs, where only 67 out of 500 runs reached the global minimum after being initialized at random positions in the PES. However, considering that the grid contains approximately 15 minima, this is not unexpected as random starting positions are likely to be closer to other local minima.

The Bayesian optimization traces in Figure 3 show that the GP based search can still identify the global minimum efficiently as long as an optimal β value is used. The dependence on β reflects the balance between exploitation and exploration in the LCB acquisition function. Small values of β make the search favour points with low predicted energy and therefore promote exploitation of wells that already look favorable. This can give fast convergence when the global minimum is found early, but also increases the risk that the optimization remains in a local minimum once such a region has been identified, which is the case for $\beta = 0.5$. Large values of β instead increase the influence of the predictive uncertainty and encourage exploration of new regions of the PES. This improves the chance of visiting the global minimum at some point, but can also lead to many evaluations being spent in high energy areas and can delay or prevent convergence to the lowest minimum, which is shown in different fashions for $\beta = 3, 5,$ and 10. In the present case, the most reliable behavior is obtained for intermediate values, here $\beta = 1$ and $\beta = 2$, which provide enough exploration to discover the global minimum while still allowing the optimization to concentrate the sampling in its low energy region once it has been found.

The spatial plots in Figure 4 help to interpret this behavior for the favorable $\beta = 2$. The predictive uncertainty is low near sampled positions and especially around minima that have been revisited, while it remains high in regions that were never chosen by the acquisition function. High acquisition values appear where low predicted energy and non-zero uncertainty coincide, which is mainly in and around the wells that contain sampled points. Regions that are clearly high in energy receive low acquisition values even if they remain un-sampled. This pattern shows that the combined GP and LCB scheme focuses evaluations on low-energy regions that already contain samples, while largely ignoring un-sampled regions that are clearly high in energy.

The general GP model addresses a different goal, namely to approximate the full PES rather than only to locate the global minimum. The RMSE in Figure 5 decreases quickly for the first 60 training samples, then flattens and only decreases slowly in an approximately linear fashion until 200 samples have been used for training. This suggests that the main structure of the PES has already been captured after 60 samples and that additional data mainly slightly refine the model. This means that the optimal amount of training samples likely lies closer to 60, with minimal return after that.

Comparing the transition path in Figure 5 for the two different models used shows how the training and purpose of the models affect the results. The BO based GP is very accurate close to the global minimum, where most of its samples lie, while the general GP, trained on a broader set of points, follows the reference energy more closely along the entire path. This illustrates the strength of BO for efficiently locating global extremes, while a more general model is needed to accurately capture the full PES.

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